Thinles Dolkar

Ph.D. Candidate

Department of Chemistry, Indian Institute of Technology, Bombay, India

thinlesdolkar1@gmail.com | +91 7051131479 | Google Scholar



EDUCATION

Ph.D. in Chemistry
Department of Chemistry
Indian Institute of Technology Bombay, India
Supervisor: Prof. Arnab Dutta

M.Sc. in Chemistry *(2019)*
Indian Institute of Technology Kanpur, India
Supervisor: Prof. Vinod K. Singh

B.Sc. (Hons.) in Chemistry *(2017)*
Hindu College, University of Delhi, India

RESEARCH EXPERIENCE

Ph.D. Research: Tailoring Catalysts for Sustainable Energy Conversion and CO_2 Utilization: From Molecular Design to Functional Devices

Supervisor: Prof. Arnab Dutta | IIT Bombay Jan'21-Dec'26 (expected)

- Developed non-heme iron-based molecular catalysts for homogeneous electrocatalytic hydrogen evolution reaction (HER) and hydrogen oxidation reaction (HOR)
- Immobilized Fe-based catalysts on carbon electrodes; built and tested H_2-fuel cells demonstrating stable HOR performance
- Developed non-heme Fe molecular catalysts for photocatalytic CO_2-to-formate conversion.
- Designed and investigated transition-metal molecular catalysts for electrochemical CO_2 reduction to formate in aqueous conditions and elucidated the mechanism via in situ spectroelectrochemical techniques.
- Investigated heterobimetallic complexes for cooperative electrocatalytic hydrogen evolution.
- Designed and synthesized CoTe/NiS_2 heterostructures as a bifunctional electrocatalyst for overall water splitting, and evaluated their performance in an electrolyzer

---

MSc. Thesis Project: | Supervisor: Prof Vinod K. Singh | IIT Kanpur Aug'18-April'19

- Enantioselective Synthesis of 1,4-Dihydroquinolines

Summer Research Internship Program: | Prof. Manish Kumar | IIT Gandhinagar May'18-Jun'18

- Antibiotic resistance of isolated E.coli in the surface water of Ahmedabad
- Assessing Microplastic abundance in river sediment in Guwahati

PUBLICATIONS

- **Dolkar, T**.; Shah, N. A.; Trivedi, V.; Dastider, S. G.; Pal, M.; Kaushik, T.; Mondal, K.; Lahiri, G. K.; Dutta, A, "A Redox-Active Iron Azo-Pyridine Complex for Aqueous H2 Evolution and H2 Oxidation: From Mechanism to Devices" ***ACS Catal. 2026***, 16 (20), 19956–19971.
- **Dolkar, T**.; Biswas, R.; Pal, M.; Roy, A.; Dutta, A, "Rational Construction of NiS2/CoTe Nanostructures as Bifunctional Electrocatalysts for Overall Water Splitting" ***ACS Appl. Nano Mater. 2026***, *9* (39), 17827–17839.
- T. Kaushik, A. Chowdhury, M. Pal, S. G. Dastider, **T. Dolkar**, V. Trivedi, R. Biswas, S. Guria, V. Vishal, K. Mondal, A. Datta, A. Dutta, "Plasmonic Gold Antenna–Copper Molecular Complex Reactor Assembly for Exclusive CO2 to Formate Conversion under White-Light Irradiation*" J. Am. Chem. Soc. 2026*, 148, 6309–6323.
- N. A. Shah, **T. Dolkar**, S. Karim, J. Ishrat, C. Das, S. Das, A. Sinha Roy, K. Bhattacharyya, A. Dutta, "pH-Modulated activation of a pendant amine leading to rapid electrocatalytic H_2 production by a molecular copper complex in acidic water" ***Inorg. Chem. Front***. **2025**, *12* (20), 6178–6190.
- S. Das, **T. Dolkar**, N. A. Shah, S. Saha, A. Dutta, "Rational design of adenine-appended synthetic cobalt catalysts via click reaction for electrocatalytic hydrogen production*" International Journal of Hydrogen Energy* **2024**, *56*, 582–588.
- T. Kaushik, S. Ghosh, **T. Dolkar**, R. Biswas, A. Dutta, "Noble Metal Plasmon–Molecular Catalyst Hybrids for Renewable Energy Relevant Small Molecule Activation" ***ACS Nanosci. Au*** **2024**, *4*, 273–289.
- A. Ali, B. V. Meena, N. A. Shah, T. Kaushik, **T. Dolkar**, C. Ghoroi, A. Dutta, "Deploying a molecular copper catalyst for efficient degradation of commercial and industrial dyes under practical conditions" ***Environ. Sci.: Water Res. Technol***. **2023**, *9*, 2219–2225.

Collaborative Publications:

- S. Mishra, **T. Dolkar**, A. Pareek, R. Maji, A. Dutta, S. Ghosh, "Heavier Chalcogenate-Bridged Manganese Carbonyl Complexes: Synthesis, Reactivity, and Electrochemical Investigation" **Chemistry An Asian Journal 2025**, 20, e202401820
- S. Mishra, **T. Dolkar**, A. Pareek, R. Bonthapally, D. K. Maity, A. Dutta, S. Ghosh, "Beyond S and Se: Electrocatalytic Hydrogen Production by Tellurolate-Bridged Co(III)–Mn(I) Heterodinuclear Complexes" ***Inorg. Chem***. **2024**, *63*, 16918–16927.
- S. Mishra, A. Pareek, **T. Dolkar**, R. Bonthapally, A. K. Pathak, A. Dutta, S. Ghosh, "Hydride-Bridged Heteronuclear Complexes for Hydrogen Evolution: Electrocatalytic and Theoretical Studies" ***Inorg. Chem***. **2026**, *65*, 18196–18208.

Manuscripts Under Preparation:

- Unveiling the Potential of a Bioinspired Non-Heme Iron Complex for Sustainable CO_2-to-Formate Conversion in Aqueous Conditions (Authors: **Thinles Dolkar**, Arnab Dutta).
- Mechanistic Insights into Photocatalytic CO_2-to-Formate Conversion by Bio-Inspired Non-Heme Iron Complexes (**Thinles Dolkar**, Arnab Dutta).

PATENT

- Bio-Inspired Electrocatalytic Systems for Bidirectional Proton-Hydrogen Conversion and Fuel Cell / Electrolyzer Integration, Indian Patent Application No. **202621016756**

CONFERENCES/WORKSHOPS

- Awarded the Best Poster Award at the International Conference on Carbon Capture and Utilization (ICCCU 2025), Jawaharlal Nehru Centre for Advanced Scientific Research, Bengaluru, India, from Dec 15th to 19th Dec 2025 ('25)
- Selected for an oral presentation at the 9th International Conference of the Institute of Metal in Biology of Grenoble (IMBG2025) on Fuel Synthesis, held in Grenoble, France, from July 1st to July 4th 2025 ('25)
- Presented research work titled "Bio-inspired Catalyst Design Strategy for Green Hydrogen Production" at SABIC, Kolkata ('24)
- Participated in the workshop on "Electron Spin Resonance Spectroscopy and its Applications" organized by SAIF/CNRTS, IIT Bombay ('24)

AWARDS/HONORS

- **FELLOWSHIP AWARD:** Selected for the award of fellowship under the scheme of Maulana Azad National Fellowship for Minority Students for research in Chemical Sciences ('20)
- Qualified for Lectureship/ Assistant Professor and secured **All India Rank-93** among 53390 (CSIR-UGC NET) ('19)
- Qualified Graduate Aptitude Test in Engineering 2020 ('20)
- Awarded the Exceptional Summer Internship Stipend (top 10 students) by the **Center for Sustainable Development (CSD)** for demonstrating outstanding research capability and contribution toward CSD's sustainability goals ('18)
- Selected for the **INSPIRE Internship Program** by Deshbandhu College, University of Delhi, for exceptional academic performance in Class 10 ('12)

SKILL SETS

Electrochemical Techniques: Cyclic voltammetry (CV), linear sweep voltammetry (LSV), controlled potential electrolysis (CPE), electrochemical impedance spectroscopy (EIS), rotating ring-disk electrode (RRDE), in situ spectroelectrochemistry, fuel cell assembly and testing.

Characterization Techniques: Nuclear magnetic resonance (NMR) spectroscopy, UV-Vis spectroscopy, FT-IR spectroscopy, Raman spectroscopy, gas chromatography (GC), gas chromatography-mass spectrometry (GC-MS), liquid chromatography-mass spectrometry (LC-MS), CHNS elemental analysis, and ion chromatography.

Software: Nova (Autolab), OriginPro, TopSpin, Adobe Illustrator, ChemDraw, PyMOL, Fortran, SHELXT, Olex2, Mercury, Microsoft Excel, and PowerPoint.

Synthesis Techniques: Synthesis and purification of transition-metal complexes, hydrothermal synthesis, microwave-assisted synthesis.

TEACHING ASSISTANTSHIP

- Conducted and supervised laboratory sessions (Inorganic Chemistry Lab) for B.Tech students (60+ students), assisted in examinations, and evaluated lab reports

LEADERSHIP /EXTRA-CURRICULAR ACHIEVEMENTS

- **NSS Member**: organized health camps, blood donation camps, cleanliness, plantation drives, and sensitization programs by Hindu College, University of Delhi ('16)
- **Creative Team Member**: Creative team member of Annual Magazine "Substance" of the Department of Chemistry, Hindu College, University of Delhi ('15)
- **Volunteer TechConnect**: Volunteered at TechConnect**,** IIT Bombay, and presented a research prototype at the event ('22)

REFERENCES

1. **Prof. Arnab Dutta** (PhD Supervisor)
Professor, Department of Chemistry, Indian Institute of Technology, Bombay, India
Email: arnabdutta.iitb@gmail.com
Personal webpage: https://www.arnabdutta-bioinorganic-lab.com/pi

2. **Prof. Sundargopal Ghosh**
Professor, Department of Chemistry, Indian Institute of Technology, Madras, India
Email: sghosh@iitm.ac.in
Personal webpage: https://chem.iitm.ac.in/Faculty/ghosh/

3**. Prof. Vinod K. Singh**
Professor, Department of Chemistry, Indian Institute of Technology, Kanpur, India
Email: vinodks@iitk.ac.in
Personal Webpage: Vinod K Singh

4. **Prof. Debabrata Maiti**
Professor, Department of Chemistry, Indian Institute of Technology, Bombay, India
Email: dmaiti@chem.iitb.ac.in
Personal Webpage: dmaiti.com | DM lab | IIT Bombay, Mumbai, Maharashtra, India